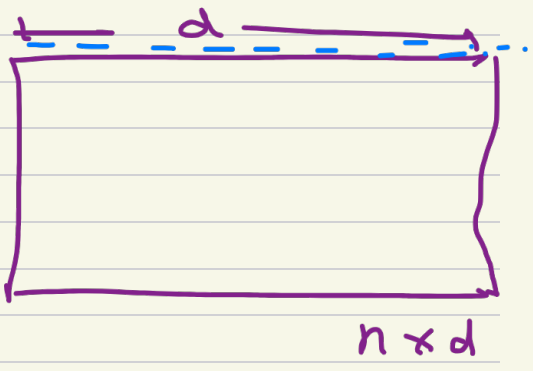
Naive Bayes - 2

1:30

* Agenda ⇒
  - → laplace smoothing
  - → Hyperparam,
  - → underflow problem
  - → feature imp.
  - → Multinomial NB

Text / emails →



of features = # of unique words

* CountVectorizer

Bernoulli

words = features

	about	bird	heard	is	the	word	you
About the bird, the bird, bird bird bird	1	5	0	0	2	0	0
You heard about the bird	1	1	1	0	1	0	1
The bird is the word	0	1	0	1	2	1	0

CountVectorizer

→ word → ? Records

→ Nos

intuition of algo

P(spam/text) or P(ham/Text)

Bayes Theorem

prior

posterior prob.

calculate likelihood

w_1 w_2 w_3 --- w_d

X conditioned on class label

(a) Simple assumption $\Rightarrow$

(b) $P(w_1 \mid y=1)$

(c) $P(w_2 \mid y=0)$

$P(w_1 \mid y=1) \times P(w_2 \mid y=1) \times \cdots \times P(w_3) \times P(w_4) =$

Hawa = hi

Sun

Spam → str.contains('free')

* [# of spam emails with free / # of spam emails]

$$P(y=1) \cdot \prod_{i=1}^{d} P(w_i \mid y=1)$$

IIRy

$$P(y=0) \cdot \prod_{i=1}^{d} P(w_i \mid y=0)$$

Training & Testing time

NB → ↓

O(n.d)

↓

O(d)

new emails

email → spam | NS

Training

$P(y=1 \mid \text{text})$

$= P(y=1) \cdot \prod_{i=1}^{d} P(w_i \mid y=1)$

$P(y=0 \mid \text{text})$

Test

↳ unknown word → X Train data

xx

absurd → was not observed during Training

X

$w_1, w_2 \ldots w_n$

$$P(y=1 \mid \text{text}) = P(y=1) \cdot \prod_{i=1}^{d} P(w_i \mid y=1)$$

Product

$P(\text{absurd} \mid y=1)$

$= \frac{0}{105} = 0$

$y=0$

$P(\text{absurd} \mid Ns) = \frac{0}{420} = 0$

entire prob $\simeq 0$

(x) fix your this

Laplace | Additive combining

Problem → new word → count find prob.

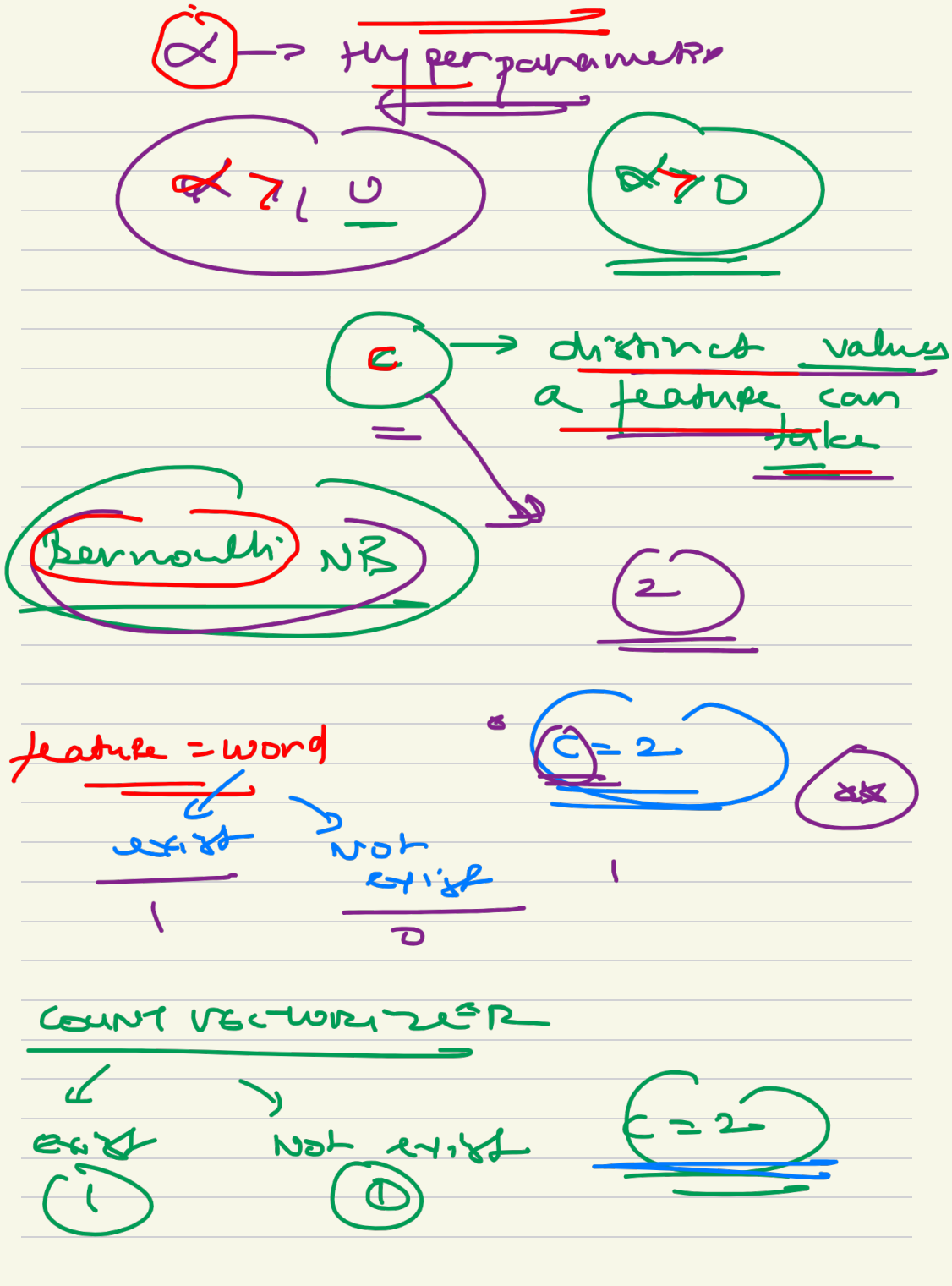
→ math collapses

Laplace smoothing ⇒

simple prob. → $P(a_j \mid y=1)$

$= \frac{n_{ji}}{n_i}$ ⇒

$= \frac{n_{ji} + \alpha}{n_i + \alpha \cdot c}$


α → Hyperparameter
α > 1 (U)
α > 0
C → distinct values a feature can take
Bernoulli NB
2
feature = word
exist
Not exist
1
0
C = 2
count vectorizer
exist
Not exist
1
0
C = 2

BOW

yes / not exist

freq. of words

C ≠ changing accordingly

α

case 1

α = 1

$$P(w' \mid y=1) = \frac{n_1' + 1}{n_1 + 2} = \frac{0+1}{100+2} \approx 0.0098$$

(0 ← n_1')

i.e

how word

or

smaller no.

maths ✗ not collapse

$\alpha \rightarrow$ Hyperparams → Tuning

$\alpha > 0$ ——

α → acts as a Regularizer

Bias vs. Trade off

overfitting → too complex

underfitting → too simple

$(\alpha = 10,000,000) \Rightarrow$

(10 Mn)

$$P(w' \mid y=1) = \frac{n_1' + 10Mn}{n_1 + 20Mn} \quad (c=2)$$

n_1 = 100

$$= \frac{6 + 10Mn}{100 + 20Mn}$$

$$\approx 0.5$$

$$= 0.499875$$

$$P(w' \mid y=0) \approx 0.5$$

$$P(\text{free} \mid y=1) = 0.1$$

$$P(\text{free} \mid y=0) = 0.5$$

$$P(\text{lottery} \mid y=1) = 0.4$$

$$P(\text{lottery} \mid y=0) = 0.4$$

win = 0.5

α → dominating

posterior

$$P(y=1 \mid \text{text}) = P(y=1) \cdot \prod_{i=1}^{d} P(w_i \mid y=1)$$

Prior → 30%

likelihood

$$P(y=0 \mid \text{text}) = P(y=0) \cdot \prod_{i=1}^{d} P(w_i \mid y=0)$$

70%

0.8 × 0.5 × 0.5 = 0.5

0 → spa

Not Spam

dumB model

→ Not a good model ✗

underfitting

10 : 8M

5 mins

Free

Offer

PL → D

D

α ↑↑ → underfitting

α ↓↓ → overfitting

GridSearch →

Course of time limit

NB

Try with 5 values

0.1, 1, 10, 100 → 0.1

Best

0.01, 0.1, 0.2, 1 → 0.1

0.05, 0.1, 0.15, 0.2 → 0.05

Random search-2

$P(y=1 \mid \text{text}) = P(y=1) \cdot \prod_{i=1} P(w_i \mid y=1)$

y=1

y=0

d=100

0.4 x 0.7 x 0.9 ...

0.1 ...

= 0.0000000000172

small

Python → os

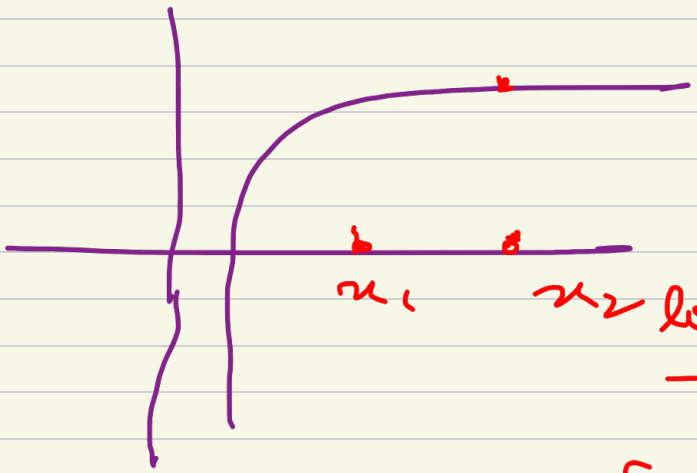
comp. arch.

add → sub

Numerical underflow problems

log on both sides

↳ monotically ↑ func^n



$x_2 > x_1$

$\log x_2 > \log x_1$

$\log[a \cdot b] = \log a + \log b$

$$\log\left[P(y=1 \mid \text{text})\right]$$

$$= \log\left(P(y=1) \cdot \prod_{i=1}^{d} P(w_i \mid y=1) \right]$$

$$\log\left(P(y=1)\right) + \sum_{i=1}^{d} \log\left(P(w_i \mid y=1)\right)$$

log of Prob.

comparison

$$\log x_2 > \log x_1$$

$y=0$

$P(y=1|text)$

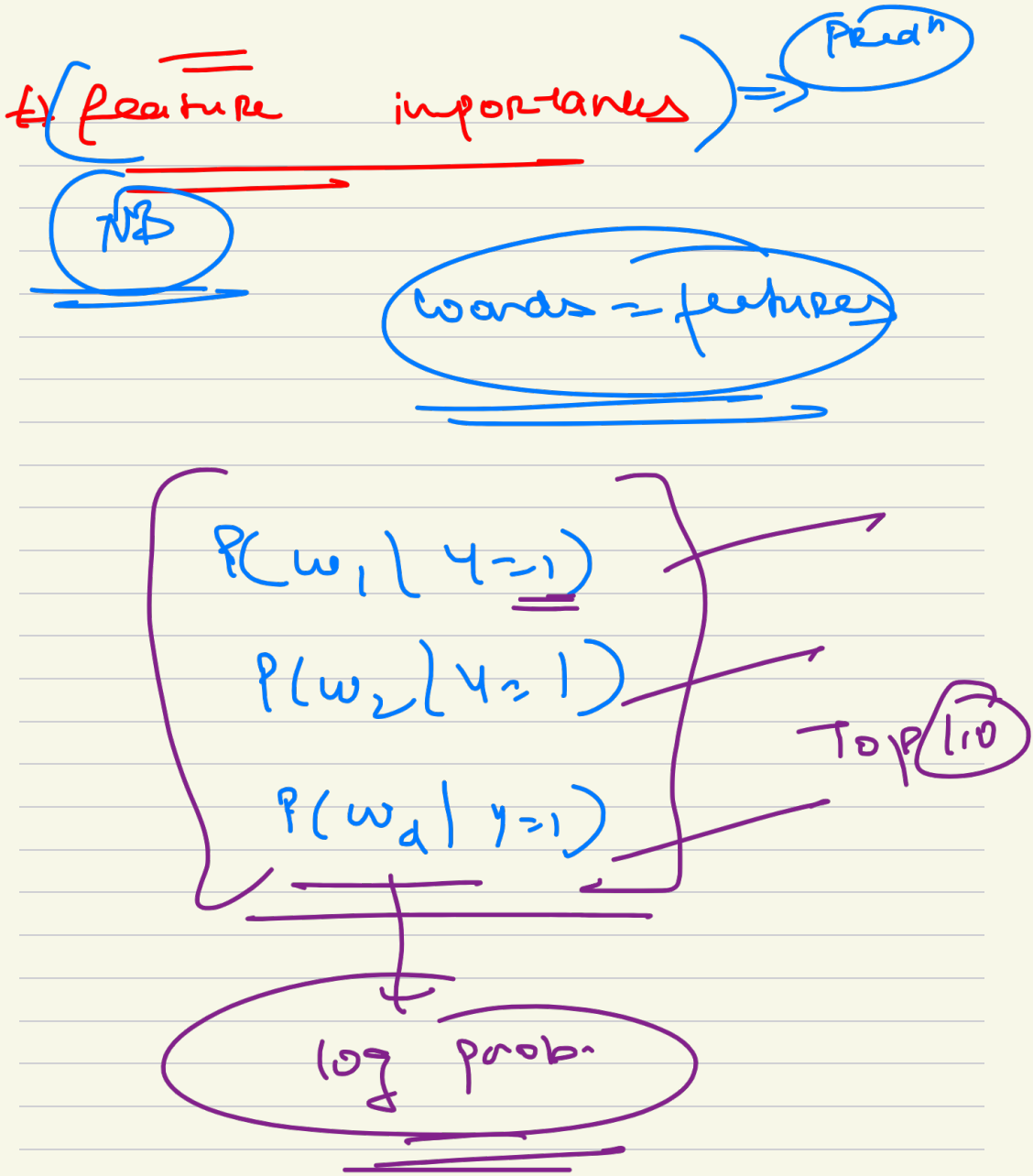
$P(y=0|text)$

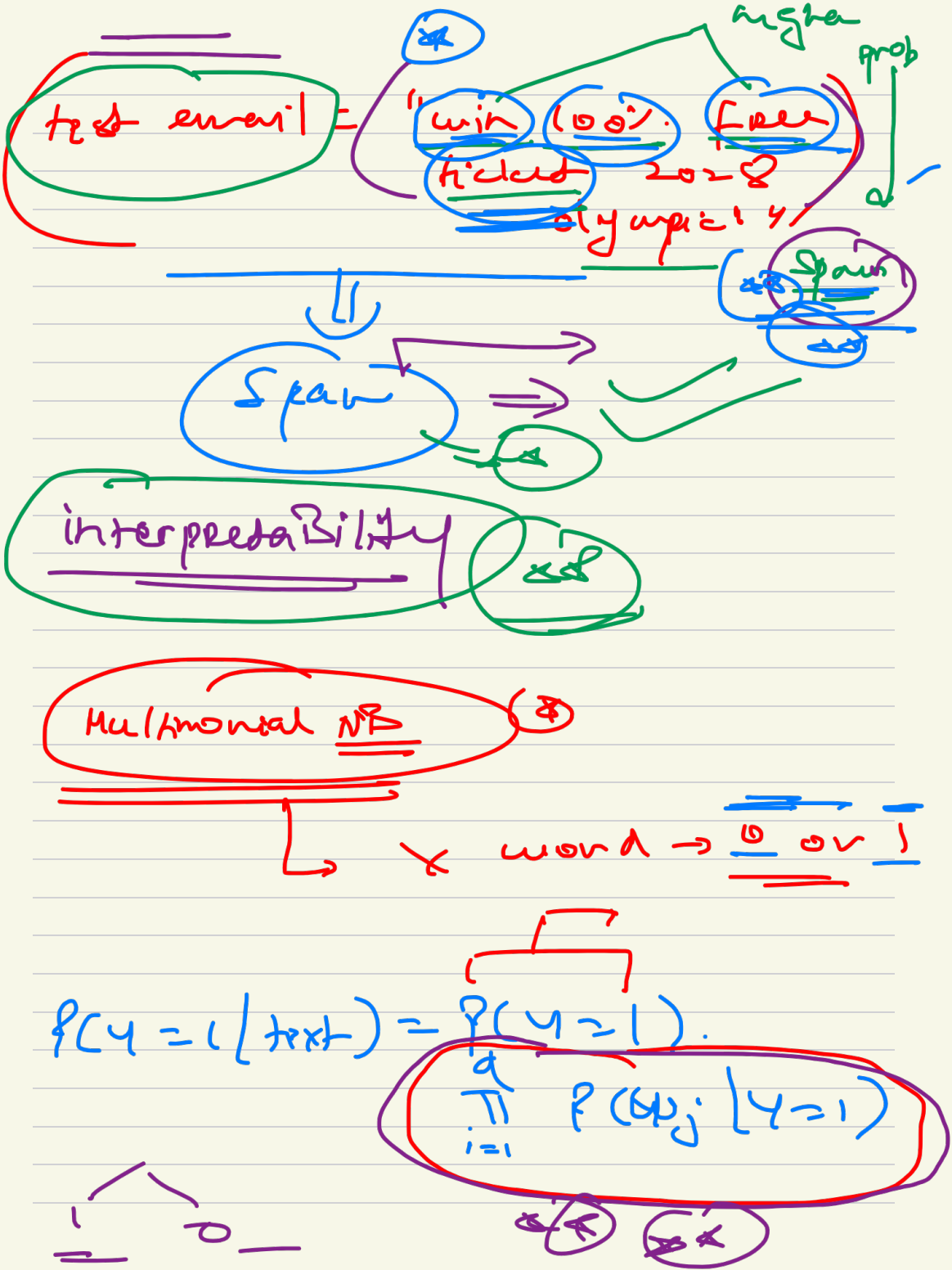
$\log(P(y=1|text))$

$\log(P(y=0|text))$

>

Bernoulli N.B →


Pred^n
f(feature importances)
NB
words = features
P(w1 | y=1)
P(w2 | y=1)
P(wd | y=1)
Top 10
log prob.


text email
"win" "00" "free"
ficlad 20-8
olympic 4/
algha
prob
Spam
Spam
interpredaBility
Multinomial NB
x word → 0 or 1
P(y=1|text) = P(y=1) ∏ P(Wj|y=1)
i=1
1
0

Vectorizer

Multinomial

	about	bird	heard	is	the	word	you
About the bird, the bird, bird bird bird	1	5	0	0	2	0	0
You heard about the bird	1	1	1	0	1	0	1
The bird is the word	0	1	0	1	2	1	0

Non spam class

Bernoulli

↓

exist or Not

w_j = free

Multinomial

→ word free exists how many times

w_j = free

free free free

$P(w_j \mid y=1) = \frac{n_{j1}}{n_1}$

$P(w_j = 3 \mid y=1)$

$P(w_j = 0 \mid y = 1)$

Test email → FREE FREE ticket is FREE

Train

Multinomial

Bernoulli

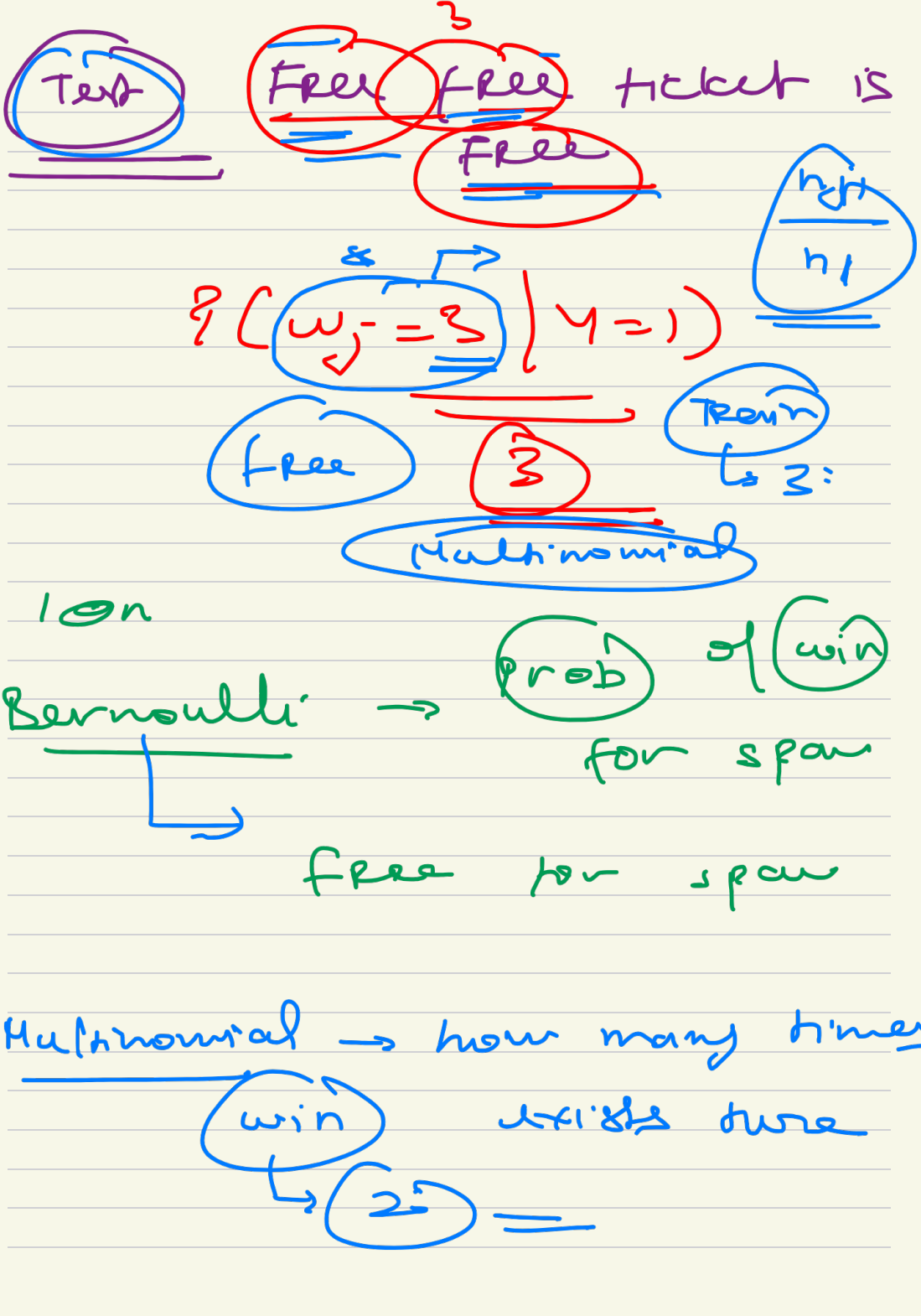
ce

v

work

verbose

more info retained


Text
Free
free
Free
ticket is
high
hi
&
P(wj = 3 | y = 1)
free
3
Train
to 3:
Multinomial
ion
Bernoulli →
prob of win
for span
free for span
Multinomial → how many times
win
exists more
25

$P(w_j = 2 \mid y = 1)$

